



Operations Research

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Solution to the Training Exercises, Sheet 6

Training Exercises

Exercise T6.1. *Knapsack Branch-and-Bound*

Problem

We consider the knapsack problem with $n = 4$ items $i \in \{1, \dots, n\}$, where item i has value $v_i \in \mathbb{N}$ and weight $w_i \in \mathbb{N}$. The backpack has total capacity of $c = 5$. The problem can be formulated as an IP as follows:

$$\begin{aligned} \max \quad & \sum_{i=1}^n v_i x_i \\ \text{s.t.} \quad & \sum_{i=1}^n w_i x_i \leq c \\ & x_i \in \{0, 1\} \forall i = 1, \dots, n \end{aligned}$$

where $x_i = 1$ means that item i is packed into the backpack. An LP relaxation is given by

$$\begin{aligned} \max \quad & \sum_{i=1}^n v_i x_i \\ \text{s.t.} \quad & \sum_{i=1}^n w_i x_i \leq c \\ & x_i \leq 1 \forall i = 1, \dots, n \\ & x_i \geq 0 \forall i = 1, \dots, n. \end{aligned}$$

Consider the following values and weights of the objects:

i	1	2	3	4
value $\mathbf{v_i}$	20	12	2	9
weight $\mathbf{w_i}$	2	2	1	2

Solve the knapsack problem by using the branch-and-bound method. Specifically, for each branching step, give an optimal solution, as well as the optimal objective value.

- Determine an optimal solution to the LP relaxation of the problem.

Hint: The LP relaxation of the general knapsack problem has the following optimal solution, which you may use without justification: Determine the value-weight ratios

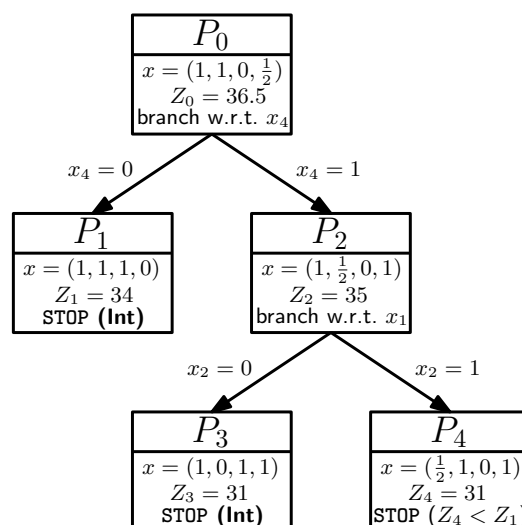
$q_i = v_i/w_i$ and index the items such that $q_j \geq q_{j+1}$ for all $j = 1, \dots, n-1$. Choose k such that $\sum_{j=1}^k w_j \leq c$, but $\sum_{j=1}^{k+1} w_j > c$. Set $x_j = 1$ for $j = 1, \dots, k$, $x_{k+1} = (c - \sum_{j=1}^k w_j)/w_{k+1}$, and $x_j = 0$ for $j > k+1$.

- b) Apply the branch-and-bound algorithm to the problem to obtain an optimal solution of the IP. In each step, state the optimal solution of the respective subproblem, as well as the optimal objective value of the intermediate solutions.

Solution

- a) The ratios are $q_1 = 10$, $q_2 = 6$, $q_3 = 2$, $q_4 = 4.5$. Using the hint, the optimal solution to the LP relaxation is thereby $x = (1, 1, 0, \frac{1}{2})$.
- b) The solution of the relaxation is given by $x = (1, 1, 0, \frac{1}{2})$. The objective value is $20 + 12 + 9/2 = 36.5$. **We branch w.r.t. x_4 .**
- **Case $x_4 = 0$:** The optimal solution can again be determined using the hint, ignoring item 4. Thus, an optimal solution is $x = (1, 1, 1, 0)$. This has the objective function value $20 + 12 + 2 = 34$. **This solution is integer, so further branching is not necessary.**
 - **Case $x_4 = 1$:** Using the hint, we obtain the optimal solution $x = (1, \frac{1}{2}, 0, 1)$. This has the objective function value $20 + 12/2 + 9 = 35$. **We branch with respect to x_2 .**
 - **Case $x_4 = 1, x_2 = 0$:** The optimal solution is $x = (1, 0, 1, 1)$. The objective value is $20 + 2 + 9 = 31$. **The solution is integer, so no further branching is done.**
 - **Case $x_4 = x_2 = 1$:** The optimal solution is $x = (\frac{1}{2}, 1, 0, 1)$. The objective value is $20/2 + 12 + 9 = 31$. **This is worse than an integer solution already found - so we branch no further.**

You can also use the tree structure to visualize the solution:



Overall, we obtain that $x = (1, 1, 1, 0)$ is an optimal integral solution.

Exercise T6.2. Gomory cuts**Problem**

For each of the following optimal tableaus of maximization problems, identify all Gomory cuts that can be derived in the first iteration.

a)

Base	Coefficients				Res.
	x_1	x_2	s_1	s_2	
z	0	0	2	1	17/3
x_1	1	0	3/2	-1/2	5/3
x_2	0	1	-2/3	4/3	4/3

b)

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
z	0	0	1	2	3	31/4
x_1	1	0	-1/2	2/3	0	5/4
x_2	0	1	3/2	-1/5	0	7/4
x_5	0	0	2/3	1	1	3

Solution

- a) The optimal solution of the tableau is $x = (5/3, 4/3)$. Since both variables are fractional, we can construct Gomory cuts from both corresponding rows.

Starting with the first constraint:

$$x_1 + \frac{3}{2}s_1 - \frac{1}{2}s_2 = \frac{5}{3}.$$

We separate each term into its integer and fractional parts, rounding each coefficient down to the nearest integer:

$$x_1 + s_1 + \frac{1}{2}s_1 - s_2 + \frac{1}{2}s_2 = 1 + \frac{2}{3}.$$

Moving all integer terms to the left and fractional parts to the right gives:

$$x_1 + s_1 - s_2 - 1 = \frac{2}{3} - \frac{1}{2}s_1 - \frac{1}{2}s_2.$$

Since $s_1, s_2 \geq 0$, we obtain the inequality:

$$x_1 + s_1 - s_2 - 1 \leq \frac{2}{3}.$$

With all variables integer-valued, the inequality simplifies to the Gomory cut:

$$x_1 + s_1 - s_2 - 1 \leq 0.$$

Alternatively, from the fractional part:

$$\frac{2}{3} - \frac{1}{2}s_1 - \frac{1}{2}s_2 \leq 0,$$

which is another valid Gomory inequality. (This could be multiplied by 6 to eliminate fractions.)

For the second constraint:

$$x_2 - \frac{2}{3}s_1 + \frac{4}{3}s_2 = \frac{4}{3}.$$

We again split terms into integer and fractional components:

$$x_2 - s_1 + s_2 - 1 = \frac{1}{3} - \frac{1}{3}s_1 - \frac{1}{3}s_2.$$

From this, we derive the Gomory cuts:

$$x_2 - s_1 + s_2 - 1 \leq 0, \text{ and } \frac{1}{3} - \frac{1}{3}s_1 - \frac{1}{3}s_2 \leq 0.$$

- b) The optimal solution is $x = (5/4, 7/4, 0, 0, 3)$. Since x_1 and x_2 are fractional, we derive Gomory cuts from their corresponding rows.

From the row for x_1 :

$$x_1 - x_3 - 1 \leq 0, \text{ and } \frac{1}{4} - \frac{1}{2}x_3 - \frac{2}{3}x_4 \leq 0.$$

From the row for x_2 :

$$x_2 + x_3 - x_4 - 1 \leq 0, \text{ and } \frac{3}{4} - \frac{1}{2}x_3 - \frac{4}{5}x_4 \leq 0$$

Exercise T6.3. Staff Scheduling

Problem

The jam factory “Waldgeist” has received several large orders for the next month and therefore need to hire additional temporary workers (temps) to meet the increased demand.

Waldgeist operates in three shifts daily (early, late and night) and there is a need for d_{st} temps on shift $s \in \{\text{early, late, night}\}$ on day $t \in \{1, \dots, 30\}$.

In response to a job posting, 20 persons $I := \{1, \dots, 20\}$ applied, all of whom are eligible.

Model an IP that you use to decide which of the temps will be employed and which will show the assignment of temps to shifts and days. After each temp has a fixed cost for the company, you want to employ as few of them as possible, but at the same time cover the demand.

However, due to legal constraints, staff scheduling must also adhere to the following rules:

- A temp must work a minimum of 5 shifts and a maximum of 15 shifts.
- A temp may work at most one shift per day.
- If a temp works the night shift, he/she may not work the morning shift on the next day.
- If a temp has a day off, they will have at least two days off in a row.

- A temp may work no more than 7 days in a row.

Hint: For the last 3 rules, it may be assumed that the temps have at least two days off both before Day 1 and after Day 30.

Solution

We denote the set of applicants by $I = \{1, \dots, 20\}$, the set of days by $T = \{1, \dots, 30\}$, and the set of shifts by $S = \{\text{early, late, night}\}$.

We introduce the following variables:

- $x_i \in \{0, 1\}$, describes if an applicant $i \in I$ is employed
- $y_{ist} \in \{0, 1\}$, describes if a temp $i \in I$ is working in shift $s \in S$ on day $t \in T$
- $w_{it} \in \{0, 1\}$, describes if a temp $i \in I$ is working (in any shift) on day $t \in T$

We want to minimize the number of temps, i.e., employed applicants:

$$\min \sum_{i=1}^{20} x_i$$

Before we formulate linear inequalities for the additional rules, we establish the relationship between the variables x , y , and w :

- If a temp works any shift on day t , she works that day:

$$3w_{it} \geq \sum_{s \in S} y_{ist} \quad \forall i \in I, t \in T$$

- If a temp works any day, they must be hired:

$$30x_i \geq \sum_{t \in T} w_{it} \quad \forall i \in I$$

Let's move on to the formulation of the rules.

Note: The variable w_{it} is not strictly necessary. Since only one shift per day is allowed, one can also simply replace w_{it} in the constraints with $\sum_{s \in S} y_{ist}$ (both expressions take values from $\{0, 1\}$).

- A temp must work a minimum of 5 shifts and a maximum of 15 shifts (*if hired*):

$$\begin{aligned} \sum_{t \in T} \sum_{s \in S} y_{ist} &\geq 5x_i \quad \forall i \in I \\ \sum_{t \in T} \sum_{s \in S} y_{ist} &\leq 15 \quad \forall i \in I \end{aligned}$$

- A temp may work no more than one shift per day:

$$\sum_{s \in S} y_{ist} \leq 1 \quad \forall i \in I, t \in T$$

- If a temp works the night shift, he or she may not work the morning shift the next day:

$$y_{i \text{ night } t} + y_{i \text{ early } (t+1)} \leq 1 \quad \forall i \in I, t \in \{1, \dots, 29\}$$

- If a temp has a day off, he or she has at least two days off in a row:

$$w_{i(t-1)} + w_{i(t+1)} \leq 1 + w_{it} \quad \forall i \in I, t \in \{2, \dots, 29\}$$

- A temp may work no more than seven days in a row:

$$w_{i(t-1)} + w_{i(t+7)} \leq 2 \left(7 - \sum_{r=t}^{t+6} w_{ir} \right) \quad \forall i \in I, t \in \{2, \dots, 23\}$$

$$\text{or} \quad \sum_{r=t}^{t+7} w_{ir} \leq 7 \quad \forall i \in I, t \in \{1, \dots, 23\}$$

Lastly, we want to make sure that there are enough temps assigned for each shift:

$$\sum_{i=1}^{20} y_{ist} \geq d_{st} \quad \forall s \in S, t \in T$$

Thus we have formulated the problem as an IP.

Self-study Exercises

Exercise S6.1. *Branch-and-Bound*

Problem

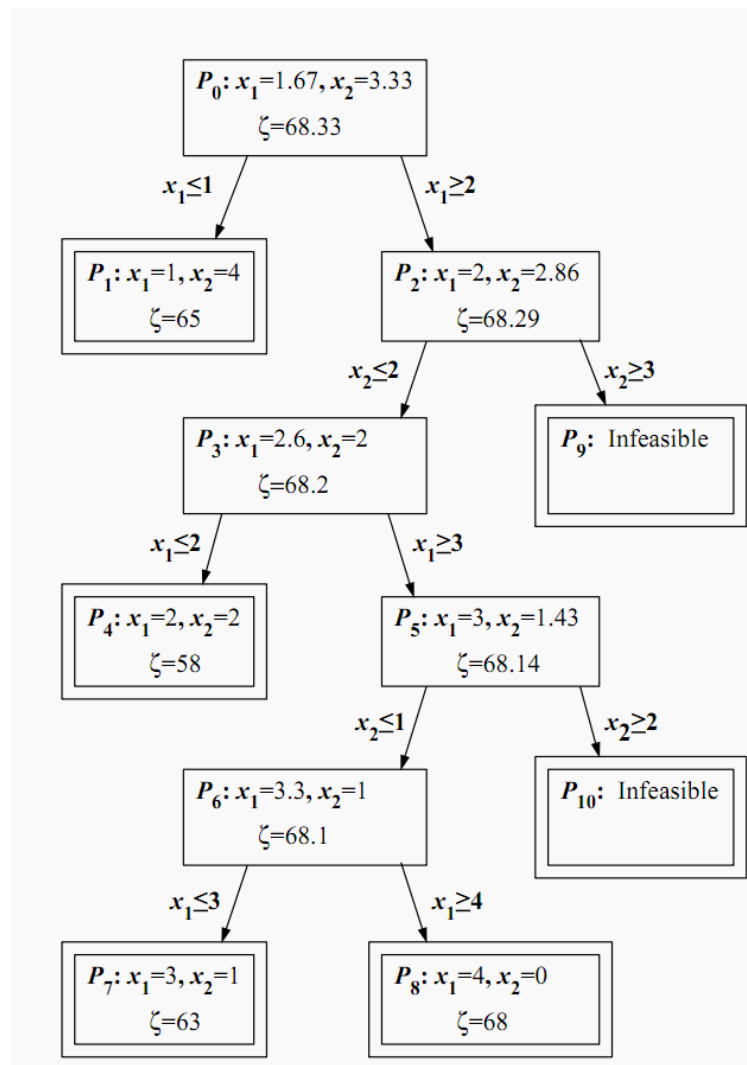
Given the following model

$$\begin{aligned} \max \quad & 17x_1 + 12x_2 \\ \text{s.t.} \quad & 10x_1 + 7x_2 \leq 40 \\ & x_1 + x_2 \leq 5 \\ & x_1, x_2 \in \mathbb{N}_0. \end{aligned}$$

Determine an optimal solution using the branch-and-bound method. (Note: To solve the (sub)problems, feel free to use Gurobi/the Internet).

Solution

Using the branch-and-bound algorithm, the following tree is obtained (ζ indicates the corresponding objective function value):



Problem

$$\begin{array}{ll}\max & 2x_1 + 3x_2 \\ \text{s.t.} & 4x_1 + 2x_2 \leq 3 \\ & x_2 \leq 3 \\ & x_1, x_2 \in \mathbb{N}_0\end{array}$$

Solution

$$\begin{array}{ll}\max & 2x_1 + 3x_2 \\ \text{s.t.} & 4x_1 + 2x_2 + s_1 = 3 \\ & x_2 + s_2 = 3 \\ & x_1, x_2, s_1, s_2 \geq 0\end{array}$$

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Base	Coefficients				Res.
	x_1	x_2	s_1	s_2	
z	0	0	3/2	0	9/2
x_2	2	1	1/2	0	3/2
s_2	-2	0	-1/2	1	3/2

From this tableau, we observe that the basic variables $x_2 = \frac{3}{2}$ and $s_2 = \frac{3}{2}$ are non-integer. Thus, we can derive two Gomory cuts.

Row x_2 :

$$2x_1 + x_2 - 1 \leq 0, \text{ and } \frac{1}{2} - \frac{1}{2}s_1 \leq 0,$$

Row s_2 :

$$-2x_1 - s_1 + s_2 - 1 \leq 0, \text{ and } \frac{1}{2} - \frac{1}{2}s_1 \leq 0.$$

For the next iteration, we include the first Gomory cut:

$$2x_1 + x_2 - 1 \leq 0.$$

Introducing a new slack variable s_3 , the constraint becomes

$$2x_1 + x_2 + s_3 = 1,$$

This leads to the following updated LP relaxation:

$$\begin{aligned}
\max \quad & 2x_1 + 3x_2 \\
\text{s.t.} \quad & 4x_1 + 2x_2 + s_1 = 3 \\
& x_2 + s_2 = 3 \\
& 2x_1 + x_2 + s_3 = 1 \\
& x_1, x_2, s_1, s_2, s_3 \geq 0
\end{aligned}$$

The resulting optimal tableau is:

Base	Coefficients					Res.
	x_1	x_2	s_1	s_2	s_3	
z	0	0	0	0	0	3
s_1	0	0	1	0	-2	1
s_2	-2	0	0	1	-1	2
x_2	2	1	0	0	1	1

This tableau corresponds to the integer solution $x = (0, 1)$. Since all variables in the solution are integer, no further Gomory cuts are required.